SEGPR: Semantic-Enhanced Geopositioning Refinement of Satellite Geographic Products by Combining Multitemporal Coarse-Accuracy Geographic Base Maps

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***Abstract*— High-resolution optical satellite geospatial products play a pivotal role in Earth observation science, as their geopositioning accuracy significantly impacts various geospatial applications. While conventional geometric correction methods rely on high-accuracy ground control points (GCPs), the global scarcity of such reference data has driven the development of GCP-free methodologies. Existing approaches, utilizing public base maps [e.g., Google Earth, shuttle radar topography mission (SRTM)] to generate coarse-accuracy GCPs (CAGCPs), predominantly employ single-temporal references, inherently constrained by the base maps' intrinsic positioning errors. Although recent attempts to manually extract multitemporal control points have partially reduced random planar errors, these methods critically neglect substantial systematic and gross errors in both elevation and planar dimensions, particularly from off-ground features in urban areas. To further improve geopositioning accuracy using multitemporal base maps, this article introduces a novel method of semantic-enhanced geopositioning refinement (SEGPR), which refines the available CAGCPs through the incorporation of semantic information and error adjustment techniques with multitemporal base maps. Specifically, this article employs a robust method to automatically capture coarse-accuracy matching control points (CAMCPs) using deep features from multitemporal base maps. Subsequently, this article innovatively explores the error characteristics of CAMCPs on both ground and off-ground features. Based on this exploration, semantic-guided random sample consensus (SGRANSAC) is introduced as a coarse correction to address elevation systematic errors, planar projection errors, and gross errors in off-ground MCPs, ensuring reliable MCP selection. In addition, the proposed elevation-augmented weight iteration (EAWI) iteratively reduces the influence of systematic elevation and random planar errors related to ground MCPs, enhancing GCP accuracy, particularly in elevation. The proposed algorithm is evaluated on geometric positioning and modeling accuracy. Experimental results across five global regions demonstrate an average geometric positioning accuracy of 1.42 m, a 55.90% improvement over state-of-the-art nonbasemap methods, and a 50.52% enhancement compared to state-of-the-art basemap methods. All corrected products achieve absolute geometric positioning accuracies better than 2 m, and modeling accuracy improves by 18.37%. The implementation of the proposed algorithm is available at https://github.com/graduate-2024/SEGPR**

***Index Terms*— Error adjustment with control points, geopositioning refinement, high-resolution optical satellite geographic products, multitemporal public geographic products, semantic-enhanced.**

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I. Introduction

SATELLITE remote sensing, with its ability to repeatedly make global observations, offers high-resolution geospatial products such as digital surface model (DSM) and digital orthophoto map (DOM). These products can capture detailed global spatial and spectral feature information and have been utilized in several significant applications, including multisource data fusion [1], [2], [3], change detection [4], [5], international disaster assessment and rescue [6]. The geometric positioning accuracy, as one of the core attributes of satellite geographic products, plays a vital role in various satellite applications [7].

The geometric positioning accuracy of satellite geographic products largely depends on the accuracy of the corresponding rational polynomial coefficient (RPC) parameters of satellite images, which are sensitive to satellite jitters, ephemeris errors, satellite clock aberration, and other factors [8], [9], [10]. Traditional methods refine RPC parameters with several high-accuracy ground control points (GCPs), thus improving the geometric positioning accuracy of satellite geographic products [11]. However, high-accuracy GCPs are difficult to obtain, especially on a global scale [12]. Consequently, it is imperative to develop geometric positioning refinement algorithms without high-accuracy GCPs.

Current methods without high-accuracy GCPs mainly utilize coarse-accuracy data with rich geospatial information as a substitute [13], [14]. Some studies [15], [16], [17], [18], [19], [20] used spaceborne laser data to improve the geometric positioning accuracy of satellite images, but the planar accuracy of

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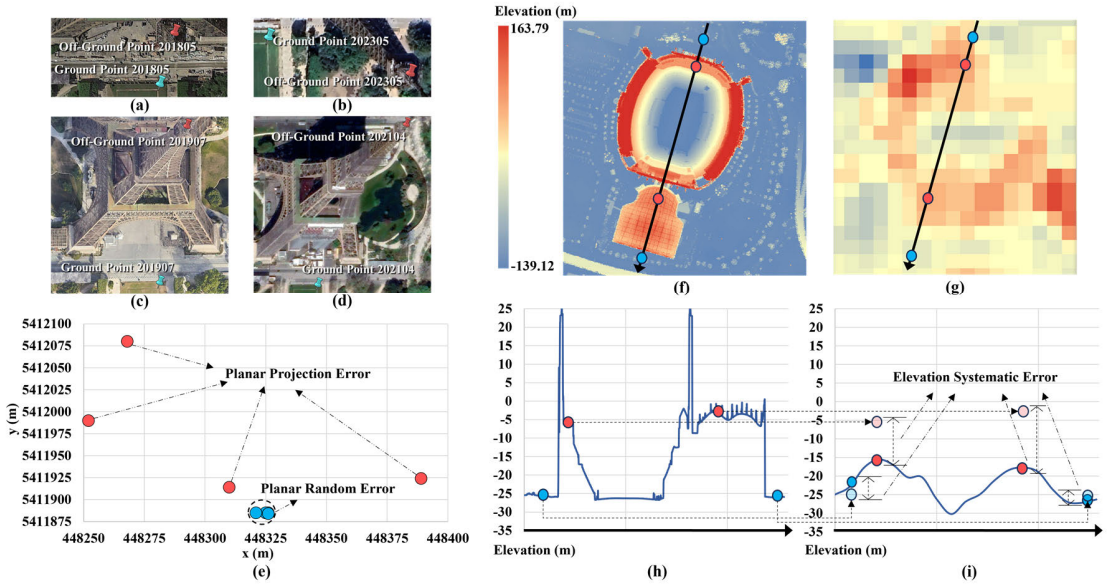


Fig. 1. In the horizontal direction, (a)–(e) display multitemporal Google Earth base maps and scatter plots representing the random errors of ground points (blue points) and the projection errors of off-ground points (red points). Elevation profiles (f)–(i) illustrating the systematic elevation errors between multitemporal base maps for ground points and off-ground points. (f) and (h) present the LiDAR ground truth, while (g) and (i) show the corresponding SRTM base maps.

such control data is relatively low. In addition, some studies have used multiple images [7], [21], [22], [23] to generate virtual control points (VCPs) to enhance the accuracy of bundle adjustment of multiview satellite images. However, such techniques are limited by the number of satellite images and the accuracy of initial RPC parameters. Geographic base maps, which are publicly available on a global scale, offer the possibility of achieving global, coarse-accuracy GCPs (CAGCPs). Most base map-based methods utilize a single-temporal geographic base map (Google Earth, etc.) or elevation base map (EBM) (shuttle radar topography mission (SRTM), advanced land observing satellite (ALOS), etc.), and can improve geometric positioning accuracy to at most approximately 5 m [24], [25], [26], [27], [28]. Nevertheless, these methods are limited by the positioning errors of single-temporal base map (SBM) data, which contain random positioning errors on the ground. Some research focuses on planar error of CAGCPs, using global or multitemporal averaging (MA) methods to deal with random errors among multitemporal base maps, applied to a limited number of manually measured areas, and achieve improved accuracy in smooth regions such as agricultural areas and forested regions [29]. However, in urban areas, CAGCPs among multitemporal base maps not only include random errors but also significant planar projection errors [see Fig. 1(a)–(e)] in buildings as well as elevation systematic errors [see Fig. 1(f)–(i)] due to complex landcover, limiting the effectiveness of multitemporal base maps in urban settings. In addition, the manually measured method struggles to fully utilize the characteristics of multitemporal base maps as redundant observations, resulting in lower efficiency of positioning accuracy refinement at the city scale.

With the development of the free-net bundle adjustment method, the relative geometric accuracy of satellite geographic products (DSM and DOM) is well ensured [30]. Furthermore, numerous satellite-derived geographic products are generated in the absence of control data, and the original satellite data, as well as RPC parameters, are frequently not retained. Therefore, this study opts to directly refine the geopositioning accuracy of the satellite geographic products for practical application purposes, rather than directly optimizing the RPC parameters. To fully exploit the potential of multitemporal base maps in urban areas by refining CAGCPs and to maximize the capabilities of multitemporal base maps as redundant observations, this study introduces a low-cost, high-accuracy geopositioning refinement method, semantic-enhanced geopositioning refinement (SEGPR) for satellite geographic products using multitemporal public base maps. The core idea of SEGPR is to fully utilize semantic information and coarse-to-fine adjustment techniques to increase the accuracy of multitemporal base map CAGCPs. In general, the proposed method first extracts coarse-accuracy matching control points (CAMCPs) with deep features from multitemporal geographic base maps. It then introduces semantic-guided random sample consensus (SGRANSAC) as a coarse correction method, ensuring the reliable selection of CAMCPs under semantic constraints. The objective of coarse correction is to mitigate the gross errors associated with CAMCPs, as well as the planar projection errors and systematic elevation errors of off-ground matching control points (MCPs). Subsequently, elevation-augmented weight iteration (EAWI) is proposed to iteratively refine the geopositioning accuracy of CAMCPs, particularly in the elevation direction, thereby generating higher-accuracy GCPs. The purpose of fine correction is to reduce the systematic elevation errors and random planar errors present in ground MCPs. Finally, the satellite geographic products are corrected using these higher-accuracy GCPs by applying the correction model. The main contributions of this article are as follows.

This article innovatively elucidates the error patterns of CAMCPs derived from multitemporal base maps, focusing

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specifically on the errors introduced by off-ground features in both planar and elevation directions. The proposed method maximizes the utility of multitemporal geographic base maps as redundant observations by employing deep learning feature matching algorithms to extract an extensive array of CAMCPs. Furthermore, based on the explored error pattern, SGRANSAC and EAWI are proposed in coarse correction and fine correction, respectively. The refinement of CAMCPs through the integration of semantic information within a coarse-to-fine adjustment strategy effectively enhances the applicability of multitemporal base maps, especially in urban areas, achieving geopositioning accuracy of satellite-derived geographic products (DOM, DSM) to 1.42 m on average, as evidenced by experimental results across five continents.

The following sections are organized as follows. Section II provides an overview of existing geopositioning refinement methods without high-accuracy GCPs. Section III describes the proposed methodology in detail. Section IV presents the experimental results and analysis. Section V concludes this article.

II. Related Work

Traditional methods can achieve high-accuracy geometric correction by deploying high-accuracy GCPs, significantly improving the absolute geometric positioning accuracy of satellite geographic products [31]. However, obtaining high-accuracy GCPs is not only costly but also challenging to achieve on a global scale [12]. To address this issue, current research has started to use coarse-accuracy control data with geographic information as "cloud control" to replace high-accuracy GCPs obtained from field surveys [13], [14]. In general, the "cloud control" data can be categorized into spaceborne laser data, VCPs, and base map data.

In studies based on spaceborne laser data, laser altimetry (LA) data with high elevation accuracy, including ice, cloud, and land elevation satellite (ICESAT)/geoscience laser altimeter system (GLAS), ICESAT2/advanced topographic laser altimeter system (ATLAS), and Gaofen-7 LA data, are primarily used to improve the geopositioning accuracy of satellite imagery. Wang et al. [18] and Li et al. [16] used ICESAT/GLAS LA data as elevation control data to improve the elevation positioning accuracy of the Mapping-1 satellite and the Ziyuan-3 satellite to approximately 2.5 and 3 m, respectively, without changing the planar positioning accuracy. Wang et al. [19] and Zhang et al. [20] used ICESAT2/ATLAS LA data to improve the elevation positioning accuracy of Ziyuan-3 satellite imagery to approximately 1.2 and 1.8 m, respectively. Hu et al. [15] and Liu et al. [17] used Gaofen-7 LA data and corresponding footprint data to improve the absolute positioning accuracy of Gaofen-7 satellite imagery and Ziyuan-3 satellite imagery to better than 6 m and approximately 4 m, respectively. Although ICESAT/GLAS and ICESAT2/ATLAS LA data are freely available worldwide, their distribution is uneven, and the planar positioning accuracy is relatively low, at the level of 10.5 ± 4.5 m [32]. In addition, Gaofen-7 LA data have an elevation accuracy of approximately 2.2 m, and the footprint data have a planar accuracy of around 5 m [15]. However, the footprint data are not currently available for public access, making it challenging to achieve global, low-cost, high-accuracy geopositioning control data.

The methods based on VCPs are mainly used for large-scale satellite mapping, which generate VCPs by multiple-image forward intersection of matches or intersection between light rays of single pixels and given elevation planes. Sun et al. [21] improved the planar and elevation accuracy of Ziyuan-3 satellite imagery to within 6 and 5 m, respectively, by using VCPs generated through the forward intersection method. Yang et al. [7] obtained VCPs through the elevation plane, improving the planar and elevation accuracy of Ziyuan-3 imagery to approximately 3 m, with an overall geometric positioning accuracy better than 5 m. Tong et al. [22] calculated planar VCPs, enhancing the planar and elevation accuracy of Ziyuan-3 imagery to 7.70 and 4.62 m, respectively. Wang et al. [23] used VCPs from the forward intersection method for large-scale bundle adjustment of Ziyuan-3 satellite imagery, achieving absolute positioning accuracy better than 5 m in China, with planar accuracy of 2.8 m and elevation accuracy of 2.6 m in other global regions. However, the virtual-control-point-based method requires a large amount of satellite images and is constrained by the accuracy of the corresponding RPC parameters. This limitation makes it challenging in scenarios of fewer satellite images and low-accuracy RPC parameters, particularly for local region measuring and microsatellite images [33].

The methods based on base map data utilize available global base map data to obtain a large number of CAGCPs, thereby improving the absolute geopositioning accuracy of satellite geographic products using these control points. Gonçalves [25] used SRTM-DEM for image orientation of ALOS-PRISM stereo pairs without high-accuracy GCPs, resulting in DEM products with root mean square errors of 2.7 and 3.9 m in the x- and y-directions, respectively, and 2.6 m in the elevation direction. Zhang et al. [27] proposed an error adjustment method by applying control information from 90-m spatial resolution SRTM-DEM to the rational function models (RFMs) for satellite image geopositioning and improving the absolute geopositioning accuracy of Ziyuan-3 satellite images to approximately 5 m. Zheng and Zhang [28] introduced a bundle adjustment method using DEM as auxiliary data, combining RFM affine transformation models with low-accuracy DEMs (elevation accuracy: 10 m, planar accuracy: 2–3 m) to enhance the absolute positioning accuracy of Ziyuan-3 and Gaofen-1 satellite images to about 5 m. Pan et al. [26] proposed a global least squares method using reference images to improve geopositioning accuracy, utilizing Google Earth base maps to enhance the geometric positioning accuracy of Ziyuan-3 and Mapping-1 satellite images to within 10 m. Cao et al. [24] used three terrain products (SRTM GL3, ASTER GDEM, and AW3D30) as coarse control constraints in bundle adjustment, improving the planar and elevation geometric positioning accuracy of Ziyuan-3 satellite images to 2.5 and 1.5 m. However, existing methods primarily use SBM data for geopositioning refinement. SBM data tend to have relatively large random errors [34]. For instance, the planar accuracy of single-temporal Google Earth orthophotos

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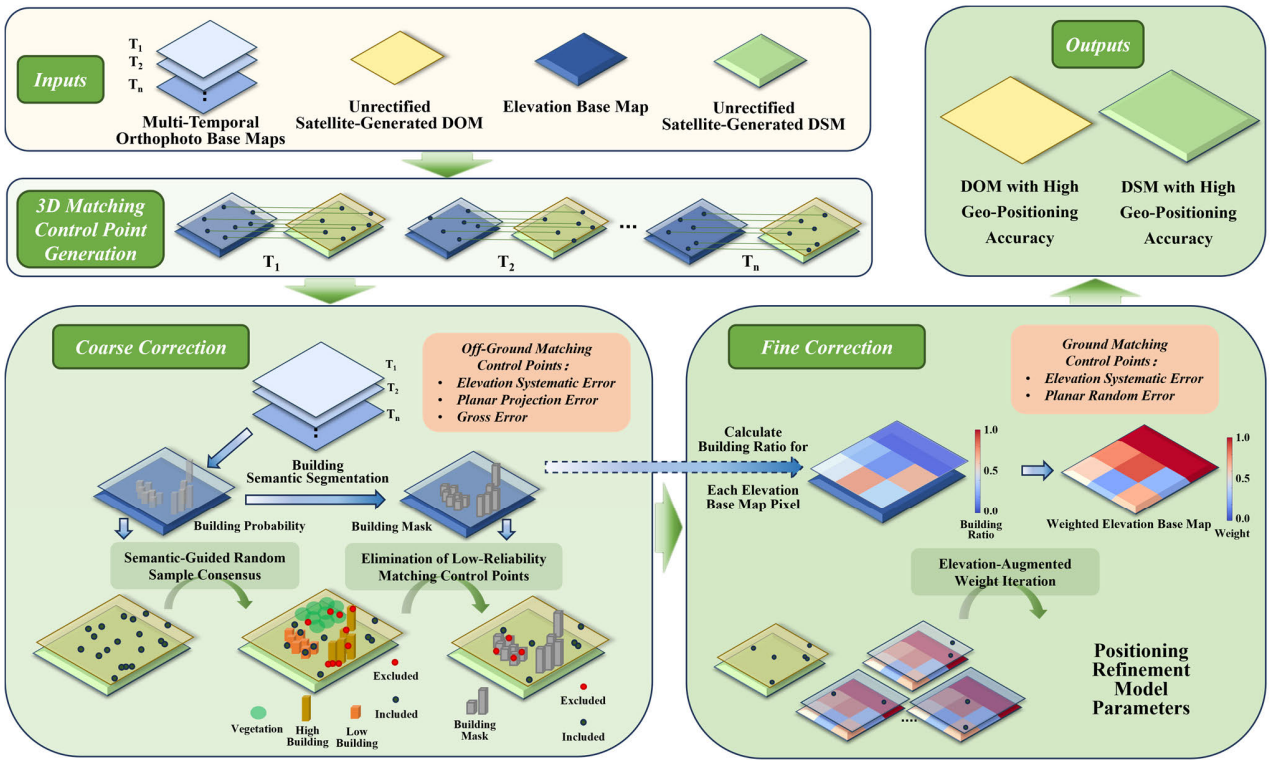


Fig. 2. Overall workflow of our proposed algorithm.

is typically 5–10 m [35], and the elevation accuracy of SRTM products is generally 6–16 m [36]. Therefore, the geometric positioning performance of such methods is severely limited by the actual accuracy of SBM data. In general, the absolute geometric positioning accuracy of satellite geographic products based on current "cloud control" methods is at best about 5 m [24], [25], [26], [27], [28], which is significantly constrained by the accuracy of SBM data.

The rapid advancement of artificial intelligence has catalyzed a paradigm shift in positioning refinement methodologies. Recent progress in this field, including but not limited to deep feature matching algorithms [37], cross-domain adaptive segmentation models (Cross-City Matter [38] and high-resolution decoupled network (HD-Net) [39]), foundation models (e.g., segment-anything model (SAM) [40] and SpectralGPT [41]), and multisource heterogeneous networks (e.g., satellite-guided bird's eye view (SG-BEV) [42]), demonstrates diverse approaches for enhancing positioning accuracy. Inspired by these developments, our method proposes a complementary integration of semantic and geometric optimization strategies. Specifically, deep feature matching networks are employed to strengthen feature representation, extracting invariant terrestrial control points from multitemporal base maps. Concurrently, semantic-aware optimization is achieved through generalizable segmentation models and subsequently incorporates semantic information to guide error consistency correction through semantic constraints. Building on SG-BEV [42]-inspired semantic weighting principles, EBM weighting is applied to achieve weighted error minimization. The proposed integration aligns with emerging trends in geospatial artificial intelligence, emphasizing the utilization of artificial intelligence models while advancing positioning refinement through explainable, semantic-driven convergence mechanisms.

Some studies have recognized random errors in multitemporal base maps and enhanced the geometric positioning accuracy of manually measured GCPs using MA. Aguilar et al. [29] first utilized multitemporal base maps with the global SRTM 30 DEM, applying a simple global averaging (GA) and outlier exclusion method at 1.25 times the standard deviation to achieve better than 2 m of planar accuracy in smooth areas covered by agricultural greenhouses. Focusing on forested areas, Wang et al. [43] proposed a MA algorithm with a 5-m sigma threshold to exclude outliers, enhancing the planar accuracy of control points to within 2 m. While these methods address random errors in smooth and forested areas in the planar direction, relying on manually selected control points, they prove less robust in urban settings due to complex terrain, gross errors, and systematic errors in built areas, particularly in the elevation direction, underscoring the necessity and significance for refinement of multitemporal base map control points.

III. Methodology

A. Overall Workflow

The comprehensive workflow of the SEGPR algorithm that integrates multitemporal geographic base maps is structured into distinct stages of refinement as depicted in Fig. 2.

Initially, uncorrected satellite-derived geographic products (DOM, DSM) along with publicly available base maps, including multitemporal orthophoto geographic base maps from Google Earth and the SRTM data, are inputted. The first stage

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involves obtaining MCPs with initially coarse accuracy by employing a deep-learning-based feature matching between satellite DOM products and the Google Earth base maps. Note that the presence of random, systematic, and gross errors in these MCPs, the workflow next applies a coarse-to-fine adjustment strategy. The coarse correction phase focuses on removing outliers, including gross errors and off-ground MCPs with elevation systematic errors and planar projection errors in building regions through the combination of the SGRANSAC algorithm and semantic building masks. The building segmentation probability is integrated as weights in sample selection in SGRANSAC. This integration effectively achieves reliable MCPs selection and robust outlier elimination.

Following the coarse correction, the subsequent phase involves meticulous fine adjustment using the innovative EAWI algorithm. This algorithm incorporates semantic weighting constraints based on elevation aspects, addressing the typical difference in the resolution of EBMs compared to orthophoto base maps. It becomes essential to assign varying weights to the MCPs to mitigate the impact of elevation systematic errors. These weights are calculated based on the proportion of buildings within mixed elevation pixels, where a single pixel may capture varying elevations due to different land cover types within the same resolution unit. Subsequently, these weights serve as initial values in an iterative process designed to refine the geopositioning refinement model parameters. This iterative process also helps to reduce the impact of random errors in ground MCPs, ensuring a more accurate and reliable geopositioning refinement.

This structured approach allows the SEGPR algorithm to significantly enhance the geometric positioning accuracy of global satellite-derived geographic products (DOM/DSM) using solely publicly accessible base map data, thereby illustrating a clear progression from initial coarse adjustments to refined positioning corrections.

B. Mathematical Model

This article utilizes satellite stereo reconstruction (SSR) software [30] to generate satellite geographic products (DSM, DOM). The free-net bundle adjustment method employed in this software ensures that the internal relative geometric positioning accuracy of the satellite geographic products is sufficiently high. However, due to the lack of high-accuracy GCPs, these satellite geographic products have low absolute geopositioning accuracy. Since the high altitude of satellite orbits, light rays of each pixel in satellite imagery are nearly parallel, approximating the geopositioning errors of satellite products as translations [44]. Therefore, this article adopts the spatial translation transformation as a geometric correction model, which assumes that the absolute geometric positioning error of the satellite geographic products consists of a 3-D spatial translation error between the object coordinate system of the satellite geographic products and the control point coordinate system

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{\text{MCP}} = \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}_{\text{Unrect}} + \begin{bmatrix} X_0 \\ Y_0 \\ Z_0 \end{bmatrix} \tag{1}$$

where $[X\ Y\ Z]^{\text{T}}_{\text{MCP}}$ is the coordinate of the MCP on base maps, $[X'\ Y'\ Z']^{\text{T}}_{\text{Unrect}}$ is the coordinate of the corresponding point on the geometrically uncorrected satellite geographic product, and $[X_0\ Y_0\ Z_0]^{\text{T}}$ is the translation parameter. The spatial translation transformation model consists of three parameters, which can be theoretically solved using only one MCP. However, due to the random errors and the systematic errors of base maps, the MCPs, especially the ones in building regions, may not be reliable. This article adopts a coarse-to-fine adjustment strategy to achieve the optimal translation parameters.

C. Generation of MCPs With Deep Features

To obtain high-accuracy adjustment accuracy of the translation parameters in (1), this article finds sufficient CAGCPs $[X\ Y\ Z]^{\text{T}}_{\text{MCP}}$ and their correspondences $[X'\ Y'\ Z']^{\text{T}}_{\text{Unrect}}$ through feature matching between the multitemporal orthophoto base maps and the satellite imagery-derived DOM. The plane coordinates of both points are extracted from the orthophoto base maps and the DOM, respectively. Since Google Earth adopts SRTM-DEM (GSD: 30 m) as the EBM, the elevation coordinates of both points are extracted from SRTM and the satellite imagery-derived DSM, respectively. The number of CAGCPs depends on the performance of the feature matching algorithm. Specifically, the construction of a cross-temporal redundant observation model through multitemporal base map data enables more effective separation of random errors and systematic biases inherent in base maps compared to single-temporal data. This spatiotemporal redundancy significantly enhances the reliability of parameter estimation in adjustment models, making the ultimate positioning accuracy improvement robust against gross errors in individual temporal phases. However, feature matching between multitemporal satellite images is a challenging problem with geometric and radiometric distortions, seasonal changes in terrestrial features, and spectral resolution differences. Fig. 3 illustrates the challenges encountered in feature matching among satellite-derived DOM and orthophoto base maps from Google Earth. The differences in spectral resolution as well as geometric and radiometric distortions between Fig. 3(a)–(d) are clearly illustrated. In addition, Fig. 3(e)–(h) vividly demonstrates the seasonal changes in terrestrial features. Traditional feature descriptors [e.g., scale-invariant feature transform (SIFT)] may achieve bad matching results in such challenging scenarios.

In the previous work [45], [46], the feature extraction network SuperPoint [47] and the feature matching network SuperGlue [37] have shown excellent performances with plenty of accurate, evenly distributed matching points in the challenging scenarios of multitemporal satellite images, even though the training dataset is street-view images [48]. As demonstrated in our previous work [45], [46], the SuperPoint + SuperGlue deep learning framework effectively acquires reliable matches in multitemporal scenarios with significant feature changes. Although prolonged intervals or seasonal variations decrease control point counts in single temporal, our strategy maintains sufficient valid control points by integrating stable features across multiple temporal. Thus, this article still utilizes SuperPoint and SuperGlue to obtain

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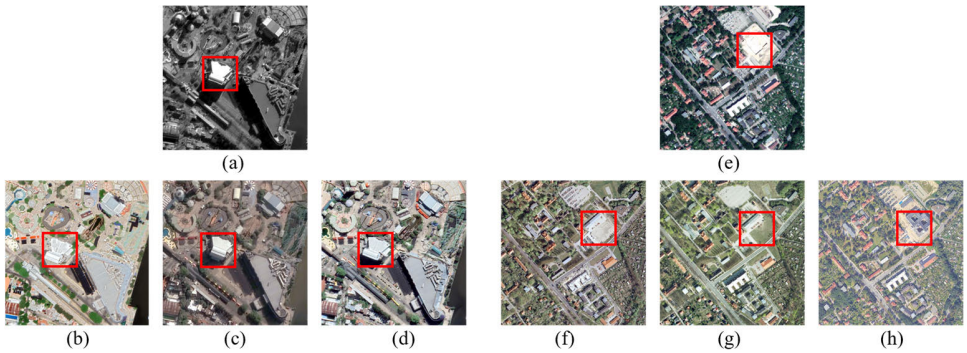


Fig. 3. Examples of challenging conditions among satellite-derived DOM and orthophoto base maps. (a) and (e) Satellite-derived DOM. (b)–(d) and (f)–(h) Orthophoto base maps.

MCPs between the base maps and the satellite imagery-derived DOM. Feature matching algorithms suppress dynamic vehicles via motion consistency constraints [37], while weak-texture water areas lack sufficient local gradients [49] to trigger feature responses. However, not all matching points are reliable, due to mismatches and the systematic projection errors of matches within buildings in base maps. A detailed analysis of the errors in base maps will be given in Section III-D.

D. Exploration of Error Patterns in Geographic Base Maps

To the best of our knowledge, this article presents the first investigation into the errors in both planar and elevation directions of MCPs generated from multitemporal orthophoto base maps and EBM. By employing deep learning feature matching algorithms, a substantial number of MCPs are obtained, revealing error patterns that are significantly more comprehensive than those of the few manually selected GCPs [29], [43]. Unlike existing studies that focus solely on planar error patterns, this investigation also includes an analysis of elevation errors, carefully distinguishing between ground and off-ground points for a comprehensive analysis. This approach provides crucial support for the design of the proposed algorithm in Sections III-E and III-F of this article.

To effectively reveal the error patterns in multitemporal orthophoto base maps and EBMs by comparing the coordinates of checking points with the ground truth, this article selects four study regions globally: Hobart in Oceania, Potsdam and Vaihingen in Europe, and Zhuhai in Asia. A series of multitemporal Google Earth base maps from 2003 to 2022 and EBM, along with the corresponding ground truth data [50], [51], [52]. Detailed data descriptions can be found in Section IV-A. The checking points were manually measured across these base maps, resulting in a total of 52 ground points and 52 off-ground points, as shown in Fig. 4. The number of temporal instances used in multitemporal base maps varies, ranging from 5 to 46 in different areas. Among all the temporals, the counts of ground points and off-ground points amount to 755 and 729, respectively. The coordinate systems of GCPs and checking points are the WGS84 universal transverse mercator (UTM) projection for the planar coordinates and the Earth Gravitational Model EGM96 for the elevation.

Regarding GCPs, this article calculates their errors by comparing the coordinates of multiple temporal GCPs and the truth



Fig. 4. Distribution of checking points in study regions. (a) Potsdam. (b) Vaihingen. (c) Zhuhai. (d) Hobart (green: ground points; blue: off-ground points).

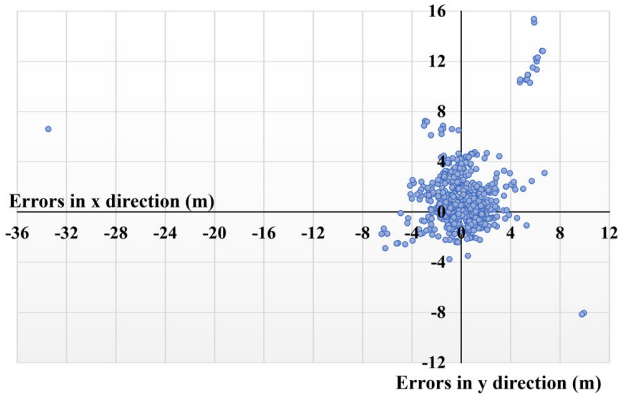


Fig. 5. Error scatter of multitemporal GCPs in x- and y-directions.

in both planar and elevation directions using (2). As illustrated in Fig. 5, the planar errors between GCPs at different time periods primarily include random errors and a few outliers. This

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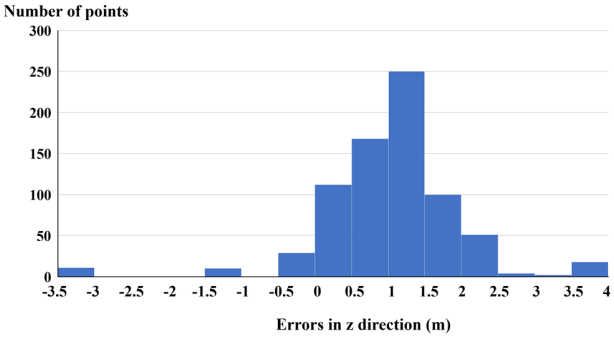


Fig. 6. Error histogram of multitemporal GCPs in z-direction.

finding aligns with previous research conclusions, indicating the presence of random errors among ground points [29], [43]. Conversely, as shown in the histogram in Fig. 6, the elevation errors of ground points across different time periods do not conform to the normal distribution of random errors. Instead, they exhibit a right-skewed distribution with a mean greater than zero, which contains systematic errors. This phenomenon is attributed to the lower spatial resolution of EBMs compared to orthophoto base maps. This lower spatial resolution causes mixed elevations of ground and surrounding objects within the same elevation pixel, resulting in GCPs that are slightly higher than the true values. Ideally, reliable MCPs should reflect only the ground elevation. Consequently, this leads to unreliable elevations of ground points influenced by the off-ground elevations distributed within the EBM pixels. Based on this analysis, this article proposes the EAWI method, which integrates semantic information to assess the reliability of elevation pixels. This approach effectively eliminates outliers through iterative processes and utilizes redundant observations to mitigate random errors, as detailed in Section III-F

$$\begin{cases} \text{Error}_{X_i} = X_i - X_{\text{TRUTH}_i} \\ \text{Error}_{Y_i} = Y_i - Y_{\text{TRUTH}_i} \\ \text{Error}_{Z_i} = Z_i - Z_{\text{TRUTH}_i}. \end{cases} \tag{2}$$

Here, Error_{X_i}, Error_{Y_i}, and Error_{Z_i} represent the errors in the X-, Y-, and Z-directions, respectively. X_i, Y_i, and Z_i denote the coordinates of the MCPs for a single period, while X_{TRUTH_i}, Y_{TRUTH_i}, Z_{TRUTH_i} correspond to the true coordinates of the MCPs.

As seen in Fig. 7, there are significant differences in the planar coordinates of the same feature points elevated above the ground level in different temporal base maps, particularly on tall buildings. In addition, as shown in Fig. 8, the EBM depicts relatively accurate elevations for terrain features such as mountains. However, the heights of buildings are significantly compressed and smoothed, resulting in substantial inaccuracies in the elevation values of buildings within the elevation direction.

The six new visual cases (see Fig. 8) demonstrate that EBMs forcibly fuse building and terrain elevations during geometric correction, causing significant height compression. Key examples include: 1) the Eiffel Tower in Paris (nonstudy area but a height anomaly case), where its extreme elevation

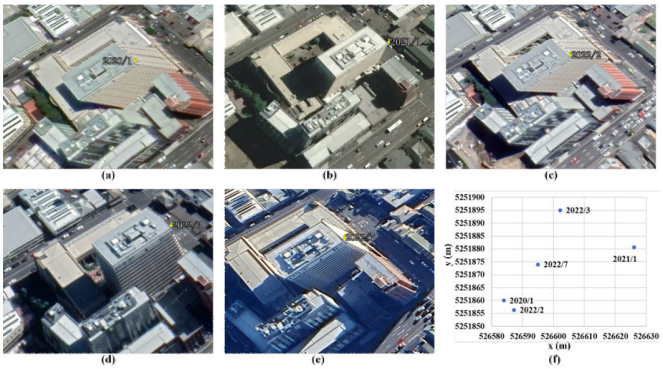


Fig. 7. Planar projection error visualization of tall buildings in Google Earth. Different temporals: (a) 2020/1, (b) 2021/1, (c) 2022/2, (d) 2022/3, and (e) 2022/7. (f) Planar coordinates.

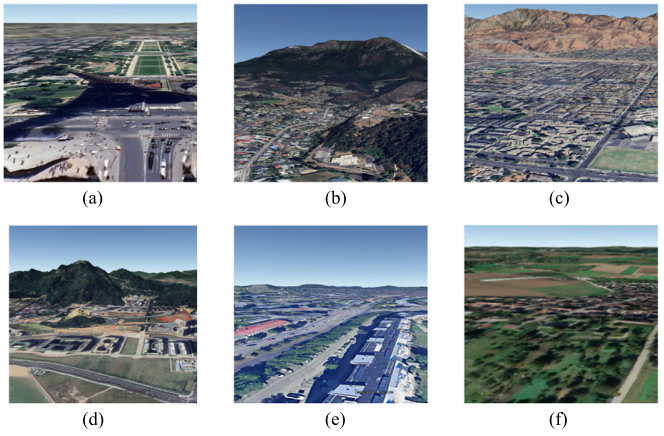


Fig. 8. Elevation systematic error visualization of built area in Google Earth. (a) Paris. (b) Hobart. (c) San Fernando. (d) Zhuhai. (e) Postdam. (f) Vaihingen.

contrast with surroundings is smoothed into a unified surface and 2) building clusters in study areas (e.g., Hobart mountains and Potsdam plains) having their independent elevations averaged with terrain data [dashed boxes in Fig. 8(b)–(f)]. Such systematic errors require correction via SGRANSAC (separating nonterrain/terrain points) and EAWI (semantic-weighted elevation adjustment).

To investigate the error patterns of off-ground control points in base map matching, this research manually measures checkpoints on buildings of different heights. This article uses the mean absolute error (MAE) metric to quantitatively analyze the coordinate errors of the selected feature points in three directions. The MAE calculation formula used to evaluate each direction is shown in the following equations:

$$\text{MAE}_{\text{PD}} = \frac{\sum_{i=1}^{n} \sqrt{\left(X_{T_i} - X_i\right)^2 + \left(Y_{T_i} - Y_i\right)^2}}{n} \tag{3}$$

$$\text{MAE}_{\text{ED}} = \frac{\sum_{i=1}^{n} \sqrt{\left(Z_{T_i} - Z_i\right)^2}}{n} \tag{4}$$

$$\text{MAE}_{\text{GPD}} = \frac{\sum_{i=1}^{n} \sqrt{\left(X_{T_i} - X_i\right)^2 + \left(Y_{T_i} - Y_i\right)^2 + \left(Z_{T_i} - Z_i\right)^2}}{n}. \tag{5}$$

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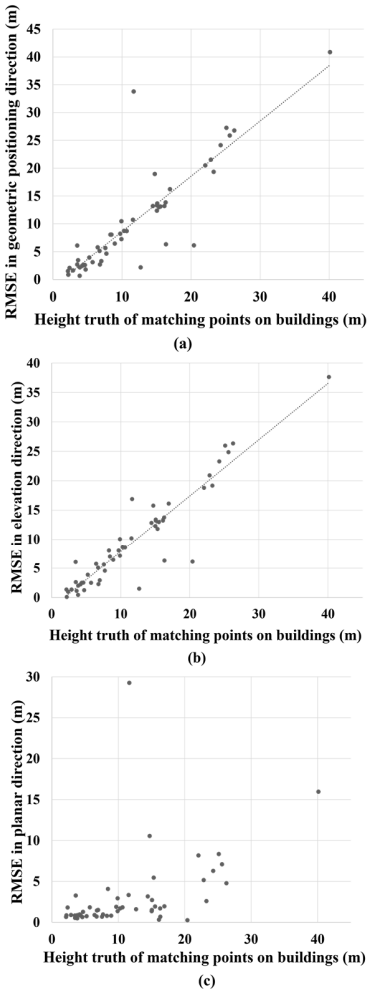


Fig. 9. Relationship between building height and errors in different directions. (a) Geometric positioning direction. (b) Elevation direction. (c) Planar direction.

Here, X_{T_i}, Y_{T_i}, and Z_{T_i} are the ith truth of 3-D coordinates of points. X_i, Y_i, and Z_i are the ith observation value of the 3-D coordinates of points.

$\mathrm{MAE_{PD}}$ is the planar accuracy of the algorithm, $\mathrm{MAE_{ED}}$ is the elevation accuracy of the algorithm, and $\mathrm{MAE_{GPD}}$ is the geometric positioning accuracy of the algorithm. Using accuracy metrics in directions of planar, elevation, and geometric positioning can, respectively, indicate the performance of algorithms comprehensively.

As seen in Fig. 9(a) and (b), the MCPs on buildings exhibit significant geometric positioning errors. In the elevation direction, the error is closely related to the height of the buildings. These elevation errors are caused by the significant differences between the Google Earth EBM (SRTM) and the actual elevations of buildings, which leads to systematic errors between EBM and DSM. On one hand, in the planar direction, the quantitative results in Fig. 9(c) indicate that the planar projection error is more random and fluctuates with the height of the buildings. This is mainly due to the random differences in the planar projection position of buildings observed from different viewing angles of satellite imagery. The findings indicate that off-ground MCPs on objects derived from multitemporal base maps exhibit both planar projection errors and elevation systematic errors. On the other hand, the geometric positioning error is positively correlated with the height of the off-ground objects. Therefore, this article proposes the SGRANSAC method to enhance the stability and robustness of the selection of MCPs with extracted building probability and mask.

E. Coarse Correction Based on Semantic-Guided Sample Consensus

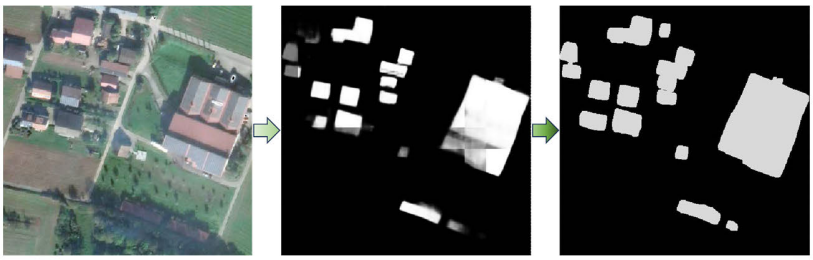
1) Extraction of Semantic Information on Multitemporal Base Maps: In urban areas, MCPs located on buildings within geographic base maps are susceptible to inaccuracies due to planar projection errors and elevation systematic errors of off-ground points. To address this issue, leveraging semantic information to refine the selection and accuracy of MCPs is crucial. A semantic-driven approach ensures that MCPs are aligned with more precise geopositioning coordinates, and the application of semantic weighting improves the elevation accuracy of MCPs. Together, these enhancements significantly boost the reliability and accuracy of geopositioning refinement algorithms, particularly in urban settings.

Multitemporal base maps, like Google Earth, provide rich spectral and texture details, facilitating the extraction of semantic information. Based on abundant semantic information, deep learning semantic segmentation algorithms can effectively extract buildings. Especially, after being trained on multiple geographic regions, they demonstrate good generalization ability, as fully validated in literature [53]. Therefore, this article utilizes RefineNet [54], pretrained on the Inria dataset [55], for semantic segmentation. The Inria dataset consists of aerial orthophoto imagery with a spatial resolution of 0.3 m, covering areas ranging from densely populated regions to small mountain towns.

For large-scale processing, this study divides the base maps into blocks with a 10% overlap. Each block is processed using a deep learning semantic segmentation algorithm to extract buildings. In this semantic segmentation method, building prediction probabilities are initially generated. These probabilities, which serve as soft constraints, guide the sampling strategy of the SGRANSAC algorithm. A threshold of 0.5 is then applied to these probabilities to create building masks. This thresholding effectively delineates areas classified as buildings, assigning a value of 1 to pixels identified as buildings and a value of 0 to nonbuilding pixels. The few segmentation errors, such as missed or incorrect segmentations, do not significantly affect the results of the proposed algorithm because the adjustment strategy effectively mitigates the impact of such errors. In this approach, semantic information primarily serves to enhance and apply soft constraints.

To address fragmentation and omissions caused by block processing, a max-pooling method is employed to merge the overlapping areas of the blocked masks, enhancing the accuracy and consistency of the segmented output. Max-pooling method can reduce edge missing probability by preserving higher building probability as high-frequency feature signals.

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(a) (b) (c)

Fig. 10. Base map and corresponding building probability and mask. (a) Base map. (b) Building probability. (c) Building mask.

The building mask generation process is shown in Fig. 10. To further refine the results and prevent misclassification of finely fragmented building probabilities, a morphological closing operation is used with a 3×3 kernel, smoothing boundaries and filling small gaps within the building areas.

Moreover, the building prediction probabilities facilitate the SGRANSAC algorithm in selecting more reliable MCPs. Subsequently, the generated building masks are used to further enhance the reliability of MCPs after the SGRANSAC process. This dual application of building probabilities and masks significantly improves the accuracy and efficiency of the georeferencing adjustments in urban environments where building structures are prevalent, ensuring a robust and accurate geospatial analysis.

2) Semantic-Guided RANSAC: Given that the initially acquired MCPs include numerous outliers, such as mismatched points and off-ground control points, this article introduces an SGRANSAC algorithm, which diverges from traditional random sample consensus (RANSAC) [56] by utilizing semantic constraints. Instead of relying solely on random selection of sample points for model fitting, as RANSAC, SGRANSAC improves the sample selection process by utilizing the semantic information of buildings. This tailored approach allows for the prioritization of more reliable ground matching points during model fitting, thereby substantially enhancing the robustness of the coarse correction process. By integrating the core principles of RANSAC with semantic guidance, SGRANSAC iteratively selects different subsets of data to find the optimal model, ensuring each selection is strategically informed by relevant semantic data to reduce the influence of outliers.

In detail, the algorithm first calculates the weights based on the building prediction probabilities obtained from the semantic segmentation algorithm at MCPs using (6). Subsequently, it constructs a cumulative weight function (7) and (8) to calculate the cumulative weight interval for each point, determining the weight range that each point falls into. Then, the algorithm performs random sampling within the range from 0 to the total cumulative weight using a uniform distribution function, resulting in a random sample weight value (9). If this weight value falls within the weight interval of a matching point (10), that point is selected as the sample for the fitting model.

a) Semantic weight calculation: For each MCP p_i, establish its sampling weight proportional to semantic reliability using

$$W_i = 1 - f(p_i) \tag{6}$$

where W_i means the weight of a point p_i, $f(p_i) \in [0, 1]$ denotes building a prediction from semantic segmentation. This inversely correlates weights with building probability, prioritizing ground points (lower $f(p_i)$) through higher weights.

b) Cumulative distribution construction: For each point p_i, calculate and construct a piecewise cumulative distribution function

$$\mathrm{CW}_i = \sum_{k=1}^{i} W_k \tag{7}$$

where W_k is the weight of a point p_k and CW_i is the cumulative weight of the point p_i.

Define the weight interval for p_i as

$$\mathrm{WI}_i = \left[\mathrm{CW}_{i-1}, \mathrm{CW}_i\right) \tag{8}$$

where WI_i is the cumulative weight interval of a point p_i. Set $\mathrm{CW}_i = 0$ by convention. This creates a continuous probability space where interval length corresponds to point weight.

Random sample using a uniform distribution sampling function within the total N matching points

$$\omega \sim U(0, W_N) \tag{9}$$

where $W_N = \sum_{i=1}^{N} W_i$ means the total cumulative weight of total N matching points. U is a function used to sample values from a given interval following the principles of uniform distribution.

The selected point satisfies

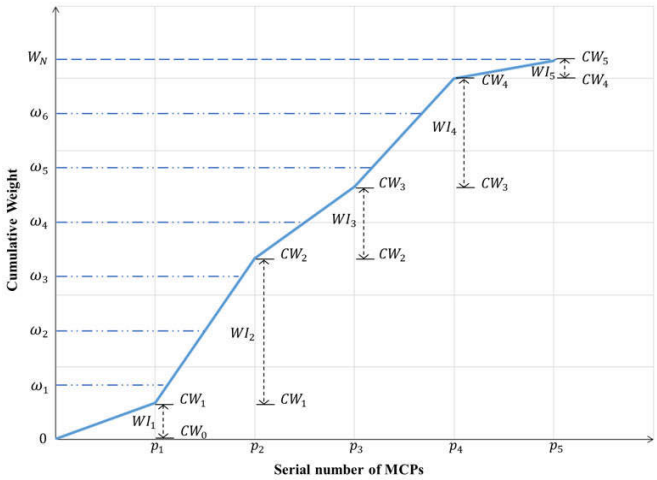
$$p^* = \{p_i | \omega \in \mathrm{WI}_i\} \tag{10}$$

where p^* is the selected sample point for the fitting model.

To enhance clarity, Fig. 11 illustrates the process described by the aforementioned formula using a simplified example. The figure presents five MCPs $p_1 - p_5$, with $\omega_1 - \omega_6$ adhering to the principle of uniform distribution sampling. As depicted, a greater weight assigned to an MCP increases its likelihood of being selected during the uniform sampling process, such as p_2 and p_4, with a longer interval length. This mechanism effectively guides the coarse correction phase by prioritizing weights derived from semantic probabilities, thereby favoring points with higher reliability for model fitting.

Thus, SGRANSAC employs a more robust and reliable sampling strategy, which is more likely to select MCPs

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Point (p_i)	Building Probability ($f(p_i)$)	Weight (W_i)	Cumulative Weight (CW_i)	Interval (WI_i)	Interval Length	Selected Probability
p_1	0.8	0.2	0.2	[0.0, 0.2)	0.2	0
p_2	0.2	0.8	1.0	[0.2, 1.0)	0.8	50% ($\omega_1, \omega_2, \omega_3$)
p_3	0.6	0.4	1.4	[1.0, 1.4)	0.4	16.7% (ω_4)
p_4	0.4	0.6	2.0	[1.4, 2.0)	0.6	33.3% (ω_5, ω_6)
p_5	0.9	0.1	2.1	[2.0, 2.1)	0.1	0

Fig. 11. Simple illustration of the sampling strategy employed by SGRANSAC.

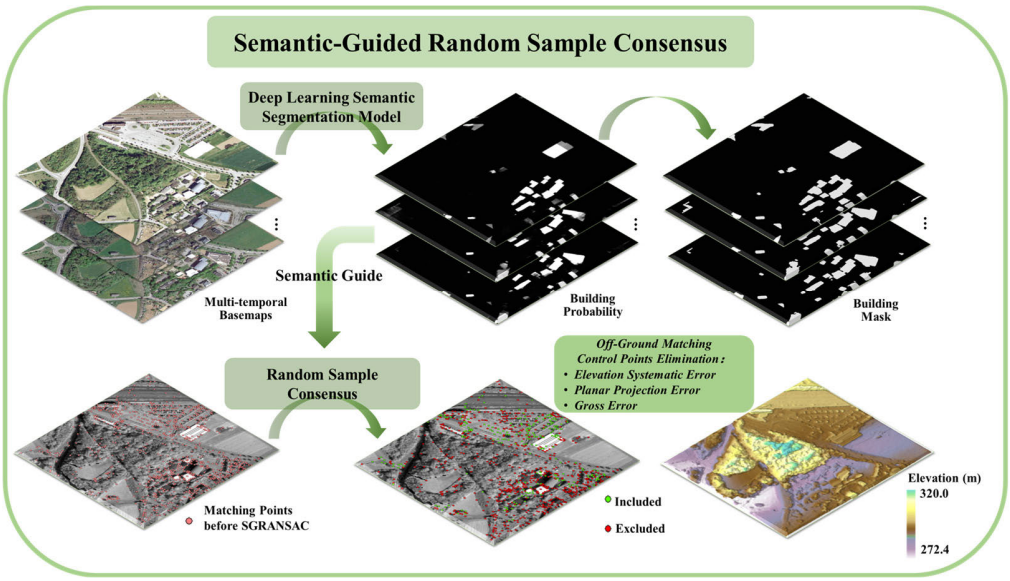


Fig. 12. Workflow of SGRANSAC algorithm, including semantic segmentation, building probability generation, and SGRANSAC.

with lower building prediction probabilities. This increases the likelihood of choosing appropriate ground MCPs and, to some extent, also accelerates the convergence of the coarse correction model. This study sets the SGRANSAC iterations at 2000, with a minimum sample size of 1 as defined in Section III-B. The model validation requires 80% of the total points as inliers, and the fitting threshold is 2.1, to be analyzed in Section IV-B1.

The workflow of the SGRANSAC is shown in Fig. 12, including semantic segmentation, building probability generation, and SGRANSAC. Following the coarse correction using SGRANSAC, mismatched control points, including MCPS on dynamic objects, water areas, at the edges of buildings, on vegetation, or relatively tall buildings with significant 3-D coordinate changes due to temporal variations, can be excluded. However, due to the threshold range inherent in SGRANSAC, it is challenging to eliminate MCPs on lower buildings. Consequently, MCPs are further refined using a building semantic mask in Section III-E3.

3) Elimination of Low-Reliability MCPs: Excluding MCPs located in building areas is crucial for refining MCPs to achieve high-accuracy georeferencing. This exclusion is facilitated by employing a binary building mask, in which building pixels are designated as 1 and nonbuilding pixels as 0. To ensure the reliability of MCPs, only those situated in areas marked as 0 in the mask are retained. This method effectively eliminates any MCPs that fall within buildings, enhancing the accuracy of the georeferencing process.

As shown in Fig. 13, after obtaining building masks from multitemporal base maps, MCPs located within these building masks are considered unreliable and thus need to be removed. MCPs on each temporal base map are filtered based on the corresponding building masks, further reducing the impact of low-rise buildings on MCPs.

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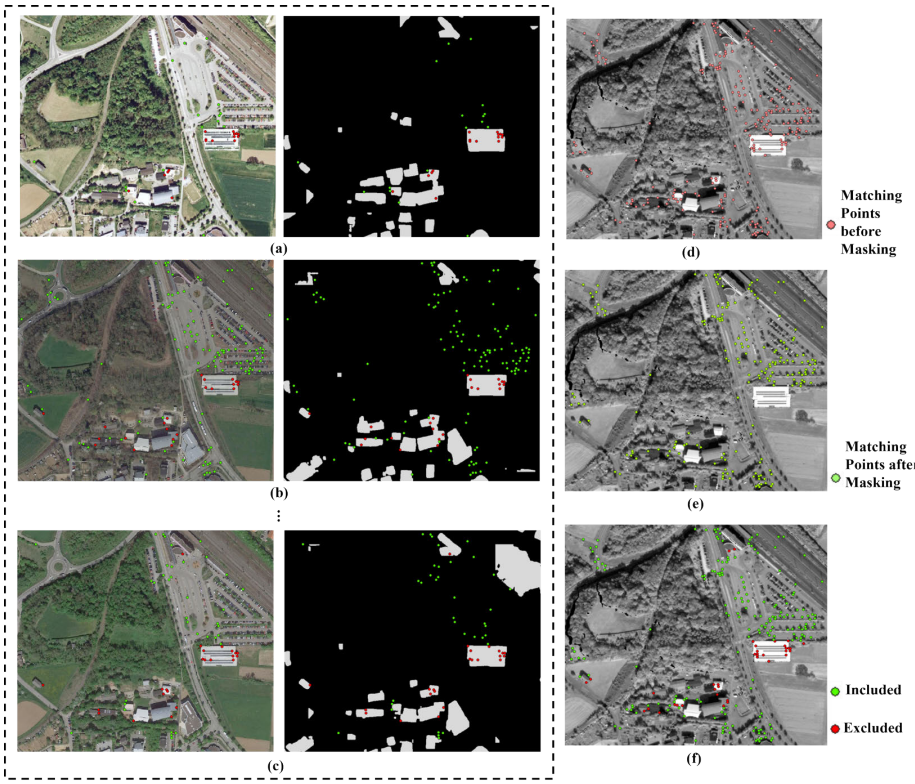


Fig. 13. Refinement of MCPs with building masks: before and after comparison on DOM. (a)–(c) Included and excluded MCPs on multitemporal base map samples and corresponding building masks. (d) and (e) Matching points before and after masking on the DOM. (f) Included and excluded MCPs on DOM through masking.

F. Fine Correction Based on EAWI

1) Elevation Weighting of MCPs: The spatial resolution of publicly available EBMs, typically ranging from 30 to 90 m, is considerably lower compared to high-resolution optical satellite geospatial products, which can reach sub-meter levels. This disparity in resolution means that a single pixel on an EBM often encapsulates a mix of ground elevation and the heights of terrestrial features, particularly buildings. Consequently, a single pixel in the EBM may include both ground and terrestrial features, which can affect the accuracy of ground MCPs within the same pixel. This imprecision necessitates the application of semantic weighting to the MCPs in the elevation direction to enhance their reliability and accuracy.

In coarse correction, the SGRANSAC algorithm has significantly reduced the number of MCPs in densely vegetated areas and on tall buildings. In addition, MCPs on low-rise buildings have been removed using building semantic masks. Consequently, most remaining MCPs are GCPs. However, the elevation accuracy of different GCPs is still affected by the EBM. Since MCPs in densely vegetated areas have mostly been eliminated, the majority of GCPs are now concentrated in areas near buildings in urban areas. Therefore, this study will next focus on the impact of buildings on the EBM.

Based on semantic information about buildings, for each pixel in the EBM, the distribution of the building mask generated from multitemporal base maps within that pixel can be calculated through geographic range consistency. For EBM pixel i, its confidence Confidence_i is as follows:

$$\text{BR} = \frac{\text{Building Pixel}_{\text{Num}}}{\text{Valid Pixel}_{\text{Num}}} \tag{11}$$

$$\begin{cases} \text{if BR} < 0.2, & \text{Confidence}_i = 1 - \text{BR} \\ \text{else}, & \text{Confidence}_i = 0. \end{cases} \tag{12}$$

$$P_{z_i} = \text{Confidence}_i. \tag{13}$$

Here, BR refers to the ratio of the building pixels among the valid pixels. $\text{Building Pixel}_{\text{Num}}$ refers to the total number of building pixels (value of 1) within the building mask of that EBM pixel. $\text{Valid Pixel}_{\text{Num}}$ denotes the total number of valid pixels (values of 0 or 1) within the building mask of the EBM pixel, excluding the impact of invalid pixels. P_{z_i} is the initial weight in the elevation direction for the ith MCP.

Utilizing the semantic information of buildings to calculate the per-pixel building ratio for the EBM provides initial weights for the elevation of MCPs in the subsequent weight iteration algorithm for fine correction, as illustrated in Fig. 14.

2) EAWI Algorithm: EAWI addresses terrain-object interference in low-resolution elevation data through a two-level weight optimization. The initial weight assignment to SRTM pixels is derived from building semantic masks, followed by iterative weight refinement to converge toward true terrain geometry. Fig. 14 demonstrates the workflow of the EAWI algorithm, including building ratio generation, elevation weight calculation, and EAWI. This framework integrates semantic awareness with geometric optimization, where the iterative weighting acts as a soft constraint to effectively avoid

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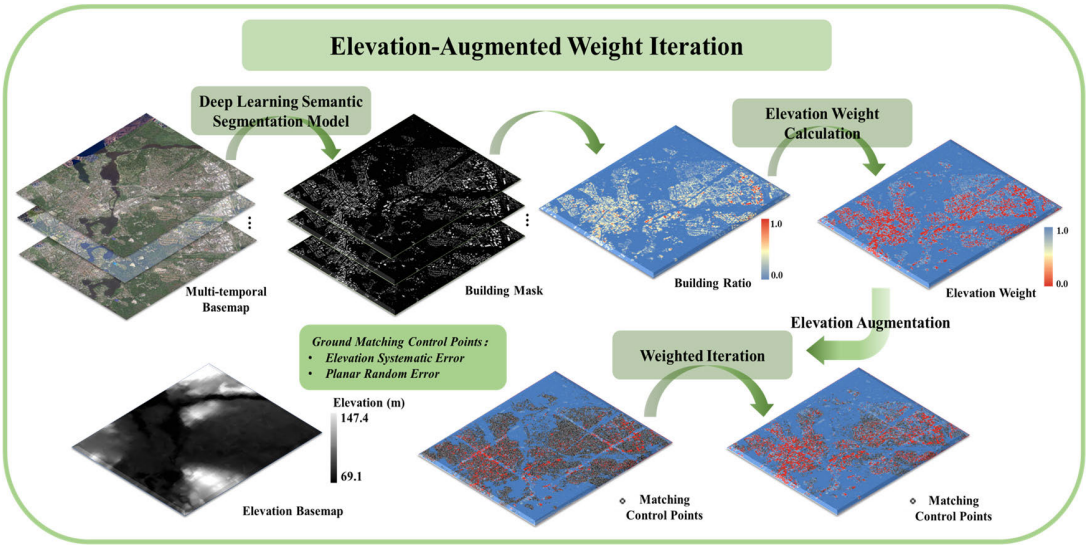


Fig. 14. Workflow of EAWI algorithm, including building ratio generation, elevation weight calculation, and EAWI.

over-reliance on low-resolution elevation weights, thereby significantly enhancing robustness in mixed-pixel regions.

Therefore, fine correction employs EAWI based on a weight iteration algorithm [57]. The initial values for the EAWI algorithm are determined first. Reliable MCPs are retained by excluding those with low reliability using SGRANSAC and building masks. In addition, initial weights for the elevation direction (z-direction) of these reliable MCPs are obtained using the EBM with confidence values. The initial weights for the planar directions (x- and y-directions) are set to 1. The initial values are then calculated using the following equations:

$$\mathrm{Avg}\left(\begin{bmatrix} X \\ Y \end{bmatrix}_{\mathrm{MCP}} - \begin{bmatrix} X \\ Y \end{bmatrix}_{\mathrm{Unrect}}\right) = \begin{bmatrix} X_{\mathrm{Init}} \\ Y_{\mathrm{Init}} \end{bmatrix} \tag{14}$$

$$\frac{P_z * \left([Z]_{\mathrm{MCP}} - [Z]_{\mathrm{Unrect}}\right)}{\mathrm{Sum}(P_z)} = \begin{bmatrix} Z_{\mathrm{Init}} \end{bmatrix} \tag{15}$$

where Avg signifies averaging, Sum indicates summation, $[X\ Y]^{\mathrm{T}}_{\mathrm{MCP}}$ and $[X\ Y]^{\mathrm{T}}_{\mathrm{Unrect}}$ represent the planar coordinates of the MCPs and the respective coordinates in the satellite geographic product DOM. P_z is the confidence level at the EBM for these control points. $[X_{\mathrm{Init}}\ Y_{\mathrm{Init}}\ Z_{\mathrm{Init}}]^{\mathrm{T}}$ denotes the initial model parameters for the weight iteration algorithm.

Next, we adopt the Institute of Geodesy and Geophysics III (IGG III) [58] robust weight function as the weighting function for our algorithm, categorizing the data into segments of credible, questionable, and gross errors. This classification facilitates the assignment of weights to each MCP throughout the iterative process. It is also critical to emphasize that in assigning weights in the elevation direction, one must consider both the weight of the MCP and the confidence level of the current pixel within the EBM. Subsequently, the following equations are presented to sequentially describe the target function, that is, the least squares criterion (16); considerations for weights in the elevation direction (17); and the IGG III weight function (18) and (19)

$$\sum_{i=1}^{n}\left(P_i v_i^2\right) = \min \tag{16}$$

$$P_i = \begin{cases} \bar{p}_i, & \text{in the direction of } x \text{ or } y \\ \bar{p}_i P_{z_i}, & \text{in the direction of } z \end{cases} \tag{17}$$

$$\bar{p}_i = \begin{cases} p_i, & |v_i/\sigma| < k_0 \\ p_i w_i, & k_0 \leqslant |v_i/\sigma| < k_1 \\ 0, & k_1 \leqslant |v_i/\sigma| \end{cases} \tag{18}$$

$$w_i = \frac{k_0}{|v_i/\sigma|}\left(\frac{k_1 - |v_i/\sigma|}{k_1 - k_0}\right)^2, \quad 0 \leqslant w_i \leqslant 1. \tag{19}$$

Here, P_i represents the weight in each direction for the ith MCP, v_i is the residual for the ith MCP, P_{z_i} is the initial weight in the elevation direction for the ith MCP, $\bar{p}_i$ is the weight in a single direction for the ith MCP, p_i is the weight in a single direction for the ith MCP after the previous iteration, w_i is the weight factor, v_i/σ is the normalized residual, and k_0 and k_1 are the corresponding critical values.

In the weight iteration algorithm, according to [59], k_0 and k_1 are set to 2.5 and 3.5, respectively, with a convergence threshold of 10^{-3}. The mean square error of prior unit weight σ is 0.25 m. The determination of the mean square error of prior unit weight σ will be discussed in detail in Section IV-B1. After convergence of the iterations, high-accuracy geometric correction model parameters can be obtained, enabling high-accuracy geometric correction of satellite geographic products (DSM, DOM).

IV. Experiments and Analysis

A. Study Areas and Data Source

This study conducts experiments in six representative areas across five continents, as shown in Fig. 15. Table I provides detailed information about each study area.

To obtain data with high absolute geometric positioning accuracy as the ground truth, this study uses high-accuracy GCPs for validation in the Hobart region [60], [61], [62]. In other areas, airborne light detection and ranging (LiDAR) data with centimeter-level absolute geometric positioning

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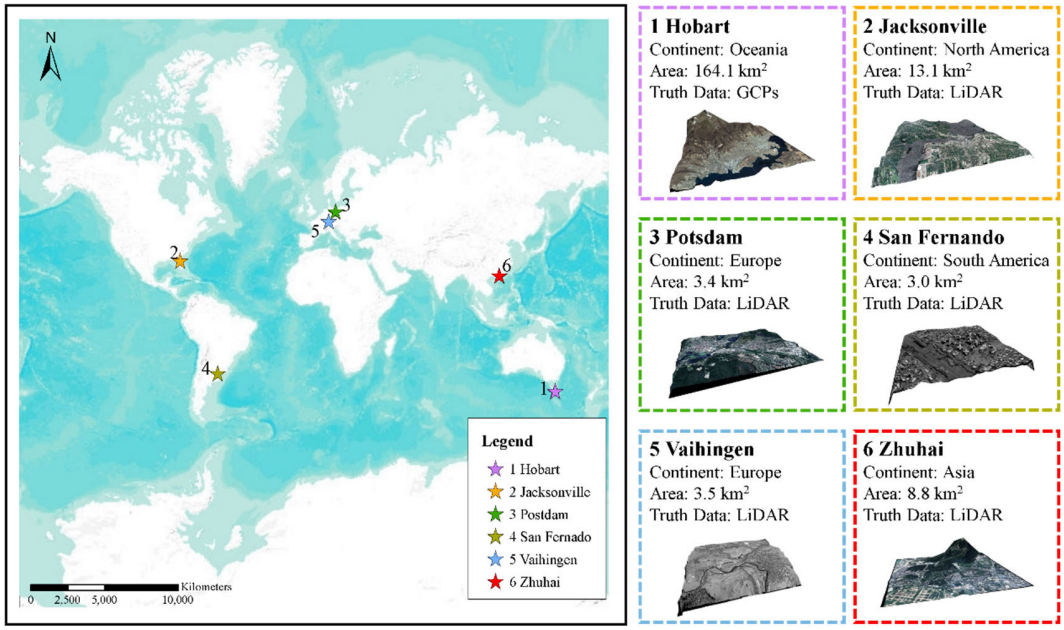


Fig. 15. Overview of study areas and data.

TABLE I
DETAILS OF STUDY AREAS AND DATA

Study Area	Satellite	Spatial Resolution (m)	Area of DSM/ DOM (km^2)	Truth	Area of Truth (km^2)
Hobart, Oceania	IKONOS	1.0	161.1	GCPs	-
Jacksonville, North America	WorldView-3	0.3	110.1	LiDAR	2.8
Potsdam, Europe	GeoEye-1	0.5	29.6	LiDAR	3.4
San Fernando, South America	WorldView-3	0.3	3.9	LiDAR	1.2
Vaihingen, Europe	WorldView-3	0.3	24.8	LiDAR	2.9
Zhuhai, Asia	WorldView-2	0.5	24.7	LiDAR	4.3

accuracy is used as the ground truth [51], [52], [63]. Both high-accuracy GCPs and LiDAR-derived control points achieve centimeter-level absolute positioning accuracy, which is significantly higher than the positioning accuracy of satellite-derived geographic products. LiDAR-derived control points and GCPs are both fundamentally high-accuracy control points used for validation.

In Section IV-B2, data from the Jacksonville region are used for hyperparameter selection, including 12 multitemporal Google Earth base maps and the SRTM-DEM elevation map. The DSM for this area, generated using LiDAR data, is obtained from open topography [U.S. Geological Survey's (USGS) 3-D Elevation Program (3DEP) data]. High absolute geometric positioning accuracy ground truth for local areas, released by the data fusion contest (DFC) in 2019, is used for high-accuracy geometric correction by manually measuring control points. After geometric correction, this section uses the high absolute geometric positioning accuracy satellite geographic product, DOM, and the corresponding LiDAR DSM as the ground truth. Next, by introducing random translational errors into the ground truth DSM and DOM, this article determines the optimal algorithm threshold through grid search. Furthermore, this article explores the impact of different base map quantities on the geopositioning refinement results.

In Section IV-B3, the proposed algorithm is applied to satellite geographic products with insufficient initial absolute geometric positioning accuracy in various regions (excluding the Jacksonville region) to perform high-accuracy geopositioning refinement, followed by a comprehensive accuracy assessment using ground truth. The base map data include multitemporal Google Earth base maps and the SRTM-DEM elevation map. The coordinate system used is the WGS84 UTM projection for the planar coordinates and geodetic height elevation system for the elevation. The preprocessing steps begin with acquiring multitemporal base maps covering DOM epochs from Google Earth Engine, ensuring temporal alignment with DOM acquisition cycles. The data are then unified into the above coordinate datums. To enhance processing efficiency, images are tiled into 1000 × 1000 to 3000 × 3000 pixels for distributed feature matching. Finally, tile-based features are fused before coarse correction to construct

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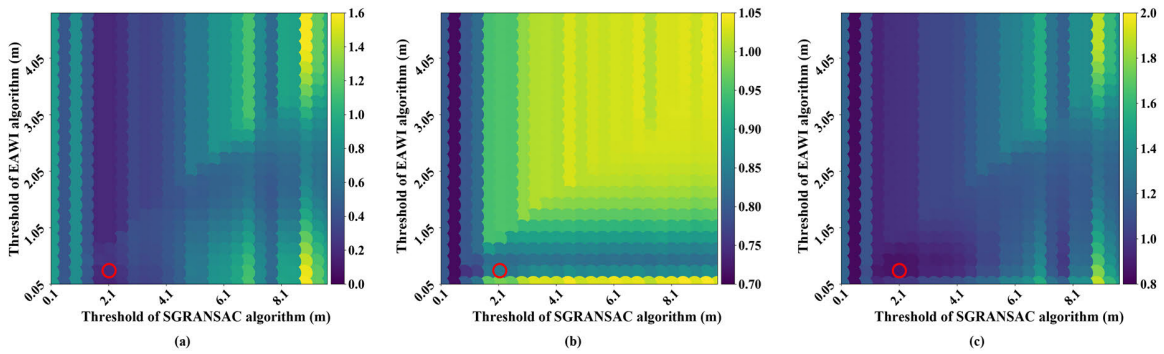


Fig. 16. Accuracy on three directions under different hyperparameter groups. (a) Elevation direction. (b) Planar direction. (c) Geopositioning direction.

global control point sets for geometric refinement. Hardware requirements include NVIDIA RTX 3090 GPU, Intel i7-13650HX CPU, 32-GB RAM, and 512-GB solid state disk (SSD) storage.

B. Results and Analysis

The experiments aim to achieve two main objectives: 1) to determine optimal thresholds for the algorithm, and explore the influence of different numbers of base maps and 2) to assess and compare the absolute geometric positioning accuracy and the modeling accuracy of satellite geographic products, both before and after applying the joint multitemporal base map geopositioning refinement algorithm. Furthermore, this article compares a method based on SBM and a method based on EBM.

1) Hyperparameter Selection: As demonstrated in Fig. 16, the threshold selection for SGRANSAC critically governs the algorithm's accuracy compared to EAWI parameters. This operational mechanism originates from SGRANSAC's dual role in eliminating gross errors while initializing geometrically valid parameters, which are derived from two globally consistent observations: spatially homogeneous SRTM elevation errors and low-rise building heights predominantly below 4 m. Following this, EAWI performs terrain-adaptive refinements within the constrained domain defined by SGRANSAC. By implementing a systematic two-stage hyperparameter optimization strategy, first establishing SGRANSAC's globally valid threshold domain (1–4 m) and subsequently identifying the optimal EAWI-SGRANSAC combination (0.25, 2.1). This parameter system achieves consistent positioning accuracy across five globally distributed test regions. These regions, including the initial calibration site, Jacksonville, span varying terrain and urban densities, thereby validating the system's strong generalization capability in diverse environments.

As Fig. 16 depicts, with a small threshold (e.g., <1 m), the sample selection of the SGRANSAC algorithm may have a weak ability to distinguish ground points and building points, since such a threshold is much lower than the actual geometric accuracy of SRTM. The SGRANSAC algorithm with a significantly small threshold will remove a large amount of outliers, including both ground points and building points, resulting in poor adjustment accuracy with insufficient point numbers. On the other hand, if the threshold is large (e.g., >4 m), the corresponding SGRANSAC algorithm will be unable to distinguish ground points and the low building points, reducing the geometric accuracy despite the robust estimation capabilities of the weighted iteration algorithm. Thus, the optimal threshold of SGRANSAC should be greater than the SRTM elevation error, while less than the height of low buildings. After analysis in Fig. 16, the optimal valid threshold interval of SGRANSAC is between 1 and 4 m.

To further improve the geometric accuracy, the choice of the mean square error of prior unit weight σ in the EAWI algorithm should be set appropriately. If set too high, the threshold makes all points appear credible, degrading the algorithm to an equally weighted average method and reducing the corresponding adjustment accuracy. On the other hand, if the value is too low, data (e.g., ground points with random errors) slightly deviating from the initial estimates are immediately marked as outliers, making the final correction parameters closely resemble the initial values provided by SGRANSAC. The optimal valid threshold interval for SGRANSAC is empirically determined as 1–4 m, and the highest geopositioning and planar accuracies are achieved at SGRANSAC $=$ 2.1 and EAWI $=$ 0.25. While this combination does not yield peak elevation accuracy, prioritizing elevation optimization (via higher thresholds) introduces spatially incoherent points that degrade planar and geopositioning accuracies. The final parameter selection, therefore, emerges from rigorous analysis of this accuracy trade-off.

The number of base maps also influences the accuracy of geopositioning refinement. This study, utilizing 12 base maps from the Jacksonville area spanning the years 2008–2022, explores the impact of base map quantity on the correction accuracy. For each quantity of base maps ranging from 1 to 12, different combinations of base maps were randomly selected three times (except when the number of base maps was 12), and the results are averaged to examine the influence of base map quantity on the correction accuracy.

The parameter tuning for basemap quantity was thoroughly validated using the Jacksonville region (110.1 km^2, 0.3-m resolution). To further corroborate the generalizability of the minimum basemap requirement, multiscale validation across two San Fernando subregions revealed consistent accuracy trends. As demonstrated in Fig. 17, there is a distinguishable decline in both geometric positioning accuracy and planar accuracy as the number of base maps increases, converging

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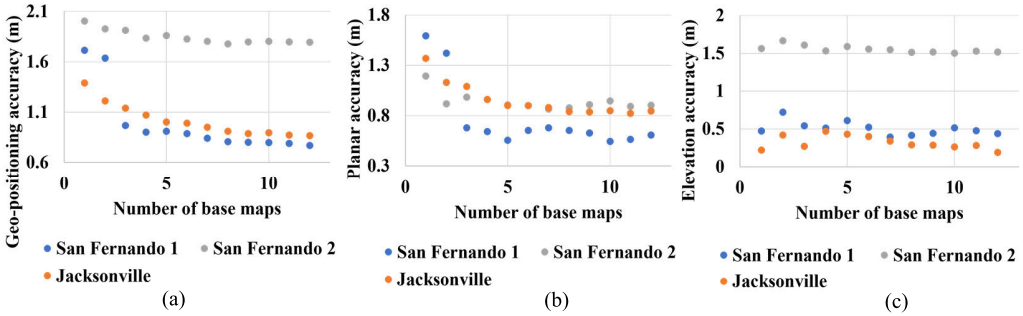


Fig. 17. Accuracy on three directions utilizing different numbers of base maps. (a) Geometric positioning accuracy. (b) Planar accuracy. (c) Elevation accuracy.

when the quantity surpasses 8. In contrast, elevation accuracy is subject to greater variability; nevertheless, it exhibits a gradual trend toward accuracy improvement with an increasing number of base maps.

2) Precision Evaluation: In this session, this article primarily compares the absolute geometric positioning accuracy across five global regions utilizing various methods: uncorrected/VCP [23], LA-based algorithm [18], GA algorithm [29], MA algorithm [43], EBM-based method [27], SBM-based method [64], and our proposed SEGPR algorithm (Ours). To highlight practical application values, this article also examines the improvements in modeling accuracy of satellite-derived products before and after corrections implemented by our proposed algorithm.

Uncorrected satellite-derived geographic products are generated using the SSR software [30]. VCP is based on the theory of VCPs [23]. The GA algorithm [29] implements a global average operation and eliminates outliers that are 1.25 times the standard deviation from 3-D MCPs. The MA algorithm [43] extends the application of automatic MCPs based on the concept of MA. This reproduction initially involves linking points between satellite-derived geographic products and multitemporal base maps, specifically searching for DSM MCPs at the same location across base maps of different temporals. Subsequently, outliers are removed from MCPs across multiple temporal dimensions using a 5-m sigma threshold. Unlike the aforementioned methods that utilize multitemporal and EBMs as inputs, the EBM method, drawing on the ideas of Zhang et al. [27], solely relies on EBMs and employs a grid iterative search for least squares fitting, optimizing the alignment between the EBMs and satellite-derived DSM products. The SBM method follows the approach described in this article, but inputs only an SBM, randomly selecting three SBMs and using their average as the result.

Absolute geometric positioning accuracy is the core metric for evaluating the effectiveness of geopositioning refinement algorithms. In terms of evaluating absolute geometric positioning accuracy, this article evaluates the accuracy of different correction methods against the ground truth (i.e., high-accuracy GCPs, LiDAR) with high absolute positioning accuracy. By manually selecting checking points on the satellite geographic products before and after correction, this article calculates the MAE in planar, elevation directions as well as the MAE of distances between the corrected points and the ground truth to assess the absolute geometric positioning accuracy. The number of checking points in Vaihingen, Zhuhai, Potsdam, San Fernando, and Hobart are 10, 10, 10, 79 (eight blocks), and 41, respectively. Spatial distribution of checking points is depicted in Fig. 18. The calculation formulas are shown in (5)–(7).

As shown in Table II, the geometric positioning accuracy of our proposed algorithm achieves an average of 1.42 m and consistently remains within 2 m across all regions, demonstrating the effectiveness and robustness of our proposed geometric correction algorithm. In terms of geometric positioning accuracy, our proposed method significantly enhanced the accuracy of origin/VCP by 55.90%. Although the LA technique demonstrates high vertical accuracy, achieving 0.52 m (109 control points) in Vaihingen and 0.74 m (12 control points) in Potsdam, it exhibits sparse spatial coverage. The methodology lacks LA data availability in other study areas and poses challenges for effective planar accuracy calibration. When compared with the state-of-the-art multitemporal base map algorithms GA and MA, our method showed improvements of 64.50% and 50.52%, respectively. It also surpassed EBM by 90.69% and outperformed SBM by 71.20%. Notably, our approach achieved marked enhancements in both planar and elevation accuracy. Specifically, in elevation accuracy, the method improved the accuracy of uncorrected products (origin/VCP) by 42.25%, which was 71.23% and 63.06% higher than that achieved by the GA and MA algorithms, respectively, and it exceeded the results of the EBM method and SBM method by 29.20% and 49.04%, respectively. In terms of planar accuracy, our approach increased the accuracy of uncorrected products (origin/VCP) by 61.22%, which was 52.34% and 31.08% better than that of the GA and MA methods, respectively, and it outperformed the EBM method by 93.28% and the SBM method by 74.88%. The comprehensive results indicate that, while current multitemporal base map algorithms have somewhat improved planar accuracy, they do not enhance elevation accuracy effectively and often yield accuracies lower than those of the uncorrected satellite-derived products. Methods based solely on EBMs, although fitting well in the elevation direction, completely overlook planar constraints, resulting in poor planar positioning accuracy. SBM methods, constrained by random errors, suffer from significant accuracy variability and instability. When significant temporal discrepancy exists between SBMs and DOM, the number

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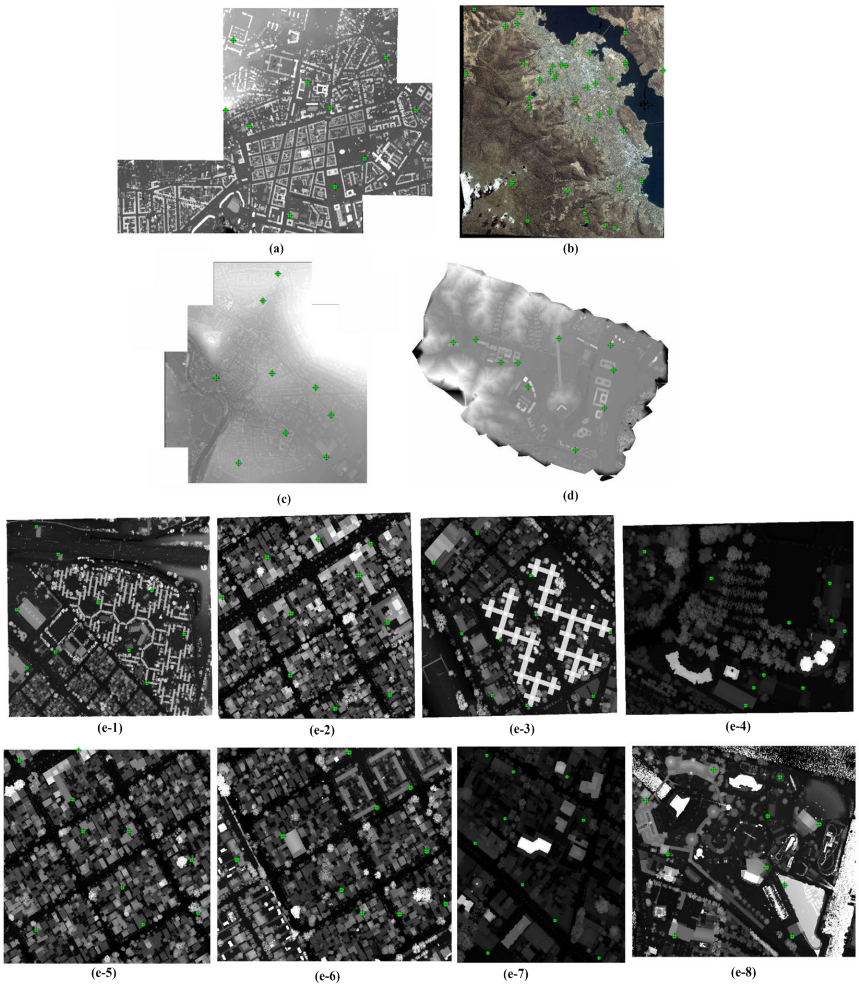


Fig. 18. Distribution of checking points in five regions. (a) Potsdam. (b) Hobart. (c) Vaihingen. (d) Zhuhai. (e-1)–(e-8) San Fernando.

of invariant matches drastically decreases due to dynamic changes of terrestrial features, resulting in insufficient valid observations. The adjustment model fails to effectively separate random and systematic errors, which ultimately causes unstable solutions.

Compared to methods that use only EBMs or SBM, our proposed approach is more robust and stable due to the additional constraints from redundant observations across multiple temporal base maps. While existing multitemporal base map methods, GA and MA, still perform adequately in regions like Vaihingen, where most areas consist of farmland and smooth terrain, they largely fail in densely built environments such as Potsdam, Hobart, San Fernando, and Zhuhai, especially in elevation direction. In these areas, our algorithm continues to effectively enhance the geometric positioning accuracy of satellite-generated products across varied terrains. The GA method primarily treats complex errors in MCPs as random, acknowledging only a minor fraction as gross errors. It utilizes a fixed standard deviation threshold to filter out gross errors, thereby neglecting systematic errors induced by off-ground terrain features affecting MCPs. In addition, the mere application of standard deviation is inadequate for substantially reducing the influence of gross errors. Expanding upon this, the MA method diminishes the impact of gross errors through the assessment of positional consistency (specifically, the Euclidean distance constraints) across different temporals for the same point, leading to enhanced results. However, it narrowly concentrates on planar random error and overlooks systematic elevation errors in base maps, rendering it ineffective in complex surface contexts.

Compared to existing multitemporal base map methods, this article innovatively explores the planar and elevation error patterns of CAMCPs in depth. Our algorithm distinctively employs a semantic-enhanced adjustment strategy tailored to correct both the planar projection discrepancies and systematic elevation errors in off-ground points, as well as the random planar and systematic elevation errors prevalent among ground points. This strategy significantly boosts the reliability and accuracy of MCPs, thereby enabling satellite-derived geographic products (DOM, DSM) to substantially enhance their geopositioning accuracy to 1.42 m on average after geopositioning refinement. Fig. 19 illustrates a significant improvement in planar accuracy within the San Fernando sample region.

On the other hand, modeling accuracy is a crucial metric for assessing the discrepancy between satellite geographic

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TABLE II
ABSOLUTE GEOMETRIC POSITIONING ACCURACY BEFORE AND AFTER CORRECTION ON THE PLANAR, ELEVATION DIRECTION, AND GEOMETRIC POSITIONING DIRECTION IN FIVE REGIONS

	Method	Study Sites					Average (m)
		Vaihingen	Zhuhai	Potsdam	San Fernando	Hobart	
Geometric positioning Accuracy (m)	VCP	1.87	2.64	2.09	5.00	4.49	3.22
	LA	1.72	/	1.47	/	/	1.60
	GA	3.03	3.83	4.62	2.57	5.97	4.00
	MA	2.32	3.39	3.89	1.51	3.25	2.87
	EBM	18.73	18.85	5.79	11.34	21.51	15.25
	SBM	5.17	2.38	2.86	3.67	10.55	4.93
	Ours	**1.16**	**1.94**	**1.55**	**1.07**	**1.36**	**1.42**
Elevation Accuracy (m)	VCP	0.81	1.73	1.58	1.39	1.58	1.42
	LA	**0.52**	/	0.74	/	/	**0.63**
	GA	2.71	3.43	4.36	2.41	1.35	2.85
	MA	2.01	2.94	3.58	1.38	1.17	2.22
	EBM	0.68	**1.04**	0.74	**0.68**	2.53	1.13
	SBM	1.96	1.16	0.52	3.03	1.20	1.57
	Ours	0.66	1.21	**0.52**	0.84	**0.85**	0.82
Planar Accuracy (m)	VCP	1.60	1.64	**1.21**	4.53	4.15	2.63
	LA	1.60	/	**1.21**	/	/	1.41
	GA	1.27	1.50	1.46	0.70	5.78	2.14
	MA	1.07	1.46	1.44	**0.47**	2.97	1.48
	EBM	18.72	18.81	5.71	11.29	21.81	15.18
	SBM	4.33	1.83	1.90	1.77	10.45	4.06
	Ours	**0.80**	**1.40**	1.41	0.54	**0.93**	**1.02**

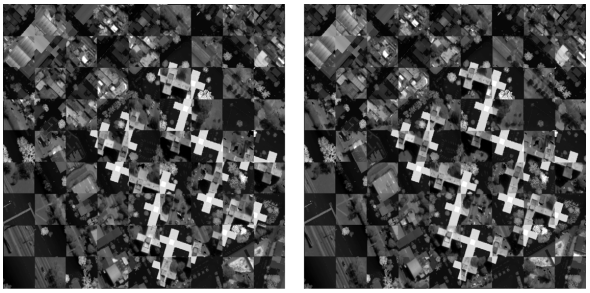


Fig. 19. Block mosaic of LiDAR DSM truth and DOM in sample region. (a) Uncorrected. (b) Corrected by multitemporal base maps using SEGPR. The block of row 1, column 1 displays the DSM truth, and each adjacent block is interleaved to display different content.

products and 3-D ground truth, impacting the practicality of satellite geographic products. In terms of modeling accuracy evaluation, this article utilizes LiDAR ground truth, known for its high absolute positioning accuracy, to assess the modeling accuracy in Vaihingen, Zhuhai, Potsdam, and San Fernando. This is achieved by calculating the MAE between the satellite geographic products and the ground truth before and after correction. The calculation formula is shown in the following equation:

$$\mathrm{MAE} = \mathrm{AVG}(|\mathrm{DSM}_{\mathrm{TRUTH}} - \mathrm{DSM}|) \tag{20}$$

where $\mathrm{DSM}_{\mathrm{TRUTH}}$ refers to the LiDAR truth, DSM refers to the DSM produced from multiview satellite images, MAE refers to the MAE, and AVG is a function that calculates the average absolute error for each region. Moreover, to avoid the influence of abnormal values on the results, the 6σ principle is used to remove the abnormal value after the error calculation.

Table III shows the significant improvement of modeling accuracy in each region after correction. On average, the overall modeling accuracy increases from 3.97 to 3.31 m on average, representing an 18.37% enhancement. Specifically, the regions of Zhuhai, San Fernando, Potsdam, and Vaihingen, respectively, experience improvements of 16.62%, 28.62%, 18.99%, and 11.11%. Furthermore, Fig. 20 presents heat maps of modeling accuracy before and after correction using our proposed method, and it demonstrates the substantial improvement in modeling accuracy. Fig. 20(a) and (d) illustrates the noticeable absolute error persisting along building edges even after geometric correction, primarily due to insufficient model accuracy at these boundaries.

3) Influence of Building Mask Error: A building mask ground-truth dataset was established through [65] and manual refinement of DOM data. Comparative experiments reveal that RefineNet-extracted masks yield more candidate points than ground truth (see Table IV) due to segmentation errors. However, after SGRANSAC geometric correction and EAWI topographic weighting, the RefineNet masks

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TABLE III
MODELING ACCURACY BEFORE AND AFTER CORRECTION IN FOUR STUDY SITES

Modelling Accuracy (m)	Study Sites				Average (m)
	Zhuhai	San Fernando	Potsdam	Vaihingen	
Uncorrected	3.97	2.97	4.74	1.89	3.32
Corrected (Ours)	**3.31**	**2.12**	**3.84**	**1.68**	**2.71**

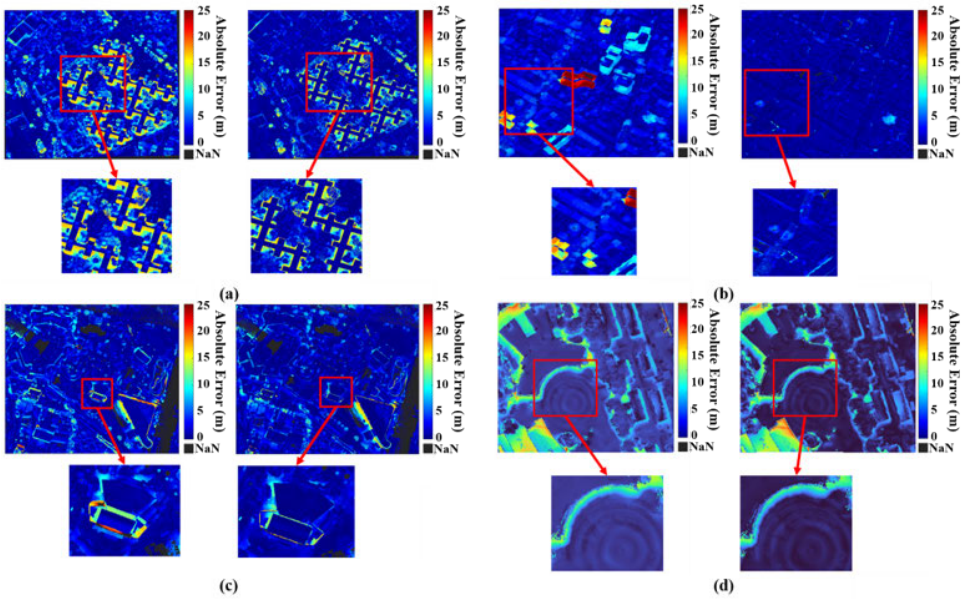


Fig. 20. Absolute error heatmap before and after correction in sample regions. (a)–(c) Sample regions in San Fernando. (d) Sample region in Zhuhai.

TABLE IV
COMPARISON OF THE EFFECTS OF USING DIFFERENT MASKS

Workflow	Truth	RefineNet-extracted masks
MCP Generation (pts)	177756	177756
SGRANSAC+mask (pts)	21850	23065
EAWI (pts)	2798	2954
Geometric positioning Accuracy (m)	1.95	1.95
Elevation Accuracy (m)	1.17	1.17
Planar Accuracy (m)	1.42	1.43

TABLE V
EFFECTS OF MORPHOLOGICAL OPERATIONS ON MCP ELIMINATION AND ACCURACY

Workflow	Truth	Dilation 3*3	Dilation 7*7	Dilation 11*11	Erosion 3*3	Erosion 7*7	Erosion 11*11
MCP Generation (pts)	177756	177756	177756	177756	177756	177756	177756
SGRANSAC+mask (pts)	21850	21037	19203	17610	22448	23156	23557
EAWI (pts)	2798	2687	2496	2304	2879	2951	2994
Geometric positioning Accuracy (m)	1.95	1.95	1.95	1.95	1.95	1.95	1.95
Elevation Accuracy (m)	1.17	1.17	1.17	1.17	1.18	1.18	1.18
Planar Accuracy (m)	1.42	1.42	1.43	1.43	1.42	1.42	1.42

exhibit merely 0.01-m planar accuracy degradation. This confirms the algorithm's robustness against segmentation errors.

Multiscale morphological transformations with $3 \times 3/7 \times 7/11 \times 11$ kernels (dilation/erosion) are applied to simulate segmentation inaccuracies (see Fig. 21). The outlier rejection

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TABLE VI
MCP CHANGES OF EACH STEP IN WORKFLOW

	Full Area	Built Area			Vegetation Area		
	MCP Number	MCP Number	Proportion (%)	Elimination Rate (%)	MCP Number	Proportion (%)	Elimination Rate (%)
MCP Generation	177756	35601	20.03	0	16603	9.34	0
SGRANSAC	24248	2397	9.89	93.27	1684	6.94	89.86
Building Mask	21917	1680	7.67	95.28	1641	7.49	90.12
EAWI	2903	107	3.69	99.70	88	3.03	99.47

TABLE VII
ACCURACY RESULTS OF EACH METHOD UNDER DIFFERENT LANDSCAPES

	Method	Study Sites				
		Vaihingen (Hill, farmland)	Vaihingen (City)	Hobart (Mountain, forest)	Hobart (Coast)	Hobart (City)
Geometric positioning Accuracy (m)	Origin	1.87	1.87	4.49	4.49	4.49
	LBM	1.74	1.71	/	/	/
	GA	3.56	3.27	5.91	5.35	6.80
	MA	1.42	2.55	4.12	2.98	3.46
	EBM	21.67	19.66	22.61	15.76	20.98
	SBM	2.23	1.95	3.54	4.16	10.64
	Ours	1.23	1.26	1.64	1.97	1.84
Elevation Accuracy (m)	Origin	0.81	0.81	1.58	1.58	1.58
	LBM	0.56	0.49	/	/	/
	GA	3.00	2.98	1.20	2.32	2.05
	MA	0.67	2.28	0.83	2.02	1.37
	EBM	0.62	0.65	2.23	1.06	1.19
	SBM	1.84	1.61	1.50	1.25	2.05
	Ours	0.71	0.76	1.26	0.58	1.14
Planar Accuracy (m)	Origin	1.60	1.60	4.15	4.15	4.15
	LBM	1.60	1.60	/	/	/
	GA	1.83	1.28	5.76	4.77	6.45
	MA	1.13	1.06	4.01	2.09	3.11
	EBM	19.65	21.66	22.49	15.72	20.94
	SBM	1.01	0.89	3.00	3.81	10.36
	Ours	0.83	0.83	0.93	1.80	1.30

TABLE VIII
STATISTICAL SIGNIFICANCE TEST OF RESULTS

	Planar accuracy	Elevation accuracy	Geo-positioning accuracy
p_value	5.74E-06	1.62E-05	1.71E-07

trajectory and final geometric accuracy metrics were systematically tracked across processing stages. Table V demonstrates that dilation operations progressively suppress control point counts. SGRANSAC + mask retains points decrease from 21 037 to 17 610 (19.4%–3.7% reduction) as kernel size increases from 3 × 3 to 11 × 11. Conversely, erosion enhances valid control points. From 3 × 3 to 11 × 11 kernels yield retained points of 2.7%–7.8% increase over the truth mask. After SGRANSAC and EAWI refinement, all experimental groups maintain planar and elevation accuracy deviations within 0.01 m, confirming tolerance to segmentation errors.

4) Changes of MCPs in Workflow: This study employs vegetation masks derived from SAM [40] and building vectors generated from literature [65] as reference data, manually

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TABLE IX
TIME CONSUMPTION OF EACH STEP IN WORKFLOW

Workflow	Zhuhai	Potsdam	Vaihingen	Hobart	San Fernando
Basemap Number	11	10	9	10	54
Area (km^2)	24.70	29.60	24.80	161.10	3.90
MCP Generation (s)	73.58	64.79	93.54	138.65	37.05
SGRANSAC (s)	0.50	5.01	0.54	7.36	0.39
Building Mask (s)	0.33	1.35	0.37	2.20	1.48
Elevation Weighting (s)	2.47	1.35	2.90	5.64	0.69
Weight Iteration (s)	0.02	0.17	0.10	0.06	0.24
Total Time (s)	76.90	72.67	97.45	153.91	39.86

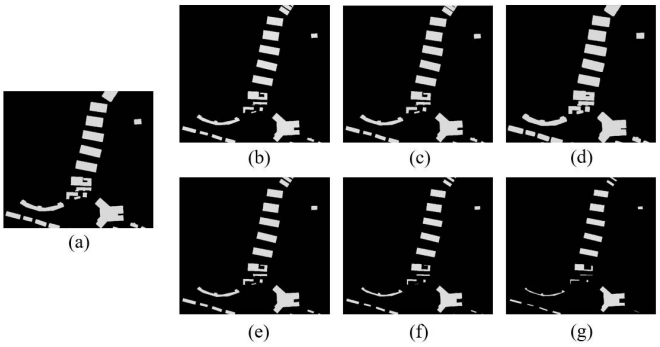


Fig. 21. Building masks after erosion and dilation with varying window sizes. (a) Ground truth. (b)–(d) Dilation results with 3 × 3, 7 × 7, and 11 × 11 windows. (e)–(g) Erosion results with 3 × 3, 7 × 7, and 11 × 11 windows.

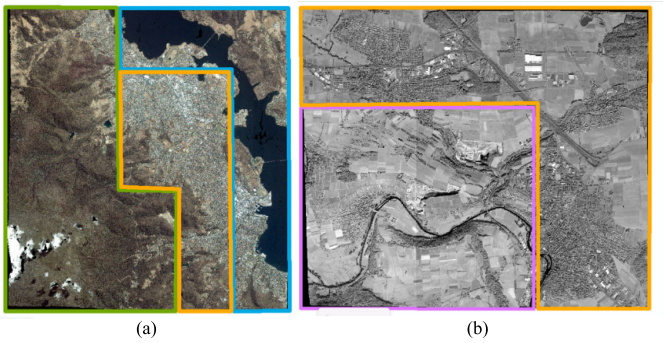


Fig. 22. Terrain classification across study regions. (a) Hobart study area. (b) Vaihingen study area. Green boxes denote mountain/forest areas, yellow boxes demarcate urban regions, red boxes indicate hill/farmland, and blue boxes represent coastal zones.

calibrated to Zhuhai's spatiotemporal features, to systematically evaluate multistage screening efficacy. As shown in Table VI, the proposed multistage framework demonstrates effective segmentation error mitigation through cascaded verification mechanisms. SGRANSAC geometric consistency screening and building mask integration achieves a 95.3% elimination rate for building control points (from 35 601 to 1680), compressing residual proportion from 20.03% to 7.67%. EAWI weight optimization further reduces building matches to 107 (weight > 0.1), with total elimination rate reaching 99.7%. For vegetation, SGRANSAC and mask processing demonstrate a 90.1% elimination rate (from 16 603 to 1641), while EAWI finalizes 88 valid points (weight > 0.1) at a 99.5% elimination rate. These results prove the SGRANSAC-mask synergy's robust outlier suppression capability and EAWI's self-adaptive optimization to spatial heterogeneity.

5) Precision Analysis of Various Terrains: Through refined terrain zoning of Hobart (161.10 km^2, mountains/forests/coastal/urban zones) and Vaihingen (hills/farmlands/urban zones), this article systematically evaluated the algorithm's performance across diverse landscapes (see Fig. 22). Key findings include: 29.31% geopositioning accuracy improvement over state-of-the-art methods in hills/farmlands, 53.67% in mountains/forests, 33.89% in coastal zones, and 36.57% average improvement in urban areas. Post-correction geopositioning accuracy remains sub-2 m (coastal: 1.97 m), demonstrating exceptional robustness even in control-point-scarce coastal terrains (see Table VII).

6) Statistical Significance Analysis: To rigorously assess the statistical significance of regional variations in positioning performance, this study implemented Kruskal–Wallis tests [1], a nonparametric rank-based method specifically designed for non-normally distributed data, to analyze accuracy differences across geographical regions in 3-D geolocation. As shown in Table VIII, the comprehensive statistical validation revealed statistically significant regional disparities in all measurement dimensions: planar positioning accuracy demonstrated substantial variations ($p = 5.74 \times 10^{-6}$, XY-axis), vertical elevation precision showed marked differences ($p = 1.62 \times 10^{-5}$, Z-axis), and most critically, composite 3-D geolocation accuracy exhibited extraordinary regional distinctions ($p = 1.71 \times 10^{-7}$). These exceptionally stringent significance levels (all exceeding conventional $p < 0.0001$ thresholds) validated the robustness of our refinement methodology across diverse geographical contexts.

7) Time Efficiency Analysis: The study quantitatively evaluates the algorithm's time performance. Experimental results demonstrate that the SEGPR method achieves geospatially scalable computational efficiency. Notably, coarse correction (SGRANSAC and mask optimization) and refinement stages (elevation weighting) account for merely 4.4% and 3.1% of total processing time, respectively, evidencing a lightweight architecture. Crucially, MCP generation dominates 92.5% of computational load, exhibiting duration dependencies on basemap quantity, resolution, and study area dimensions (see

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Table IX). While MCP complexity scales with these parameters, subsequent linear-complexity stages ensure practical deployability in large-scale scenarios. Importantly, parallel GPU acceleration (beyond our single NVIDIA 3090 implementation) presents viable optimization potential for MCP generation in real-world applications.

V. Conclusion

This study presents a significant advancement in refining the absolute geometric positioning accuracy for high-resolution optical satellite geographic products (DSM, DOM) through comprehensive error correction from redundant observations in a multitemporal base map. By integrating semantic information with robust adjustment techniques, the proposed SEGPR method, employing SGRANSAC and EAWI, can substantially improve CAMCPs' geopositioning accuracy by mitigating systematic/gross errors, particularly those induced by urban off-ground features. Experimental results across five global regions demonstrate an average enhancement of geopositioning accuracy to 1.42 m. This represents a 55.90% improvement over the state-of-the-art nonbasemap method and a 50.52% enhancement compared to the state-of-the-art multitemporal base map techniques, with absolute geopositioning accuracy better than 2 m consistently achieved. Furthermore, modeling accuracy after correction is improved by 19.23%, which better supports high-accuracy downstream geospatial applications.

While demonstrating notable advancements, the methodology's reliance on multitemporal basemap availability and quality introduces limitations, particularly in regions with large-scale water bodies/cloud coverage and scenarios involving abrupt geographical variations (e.g., seismic/volcanic terrain deformations). Although urban areas exhibit significant accuracy improvements, the approach's effectiveness in heterogeneous or rapidly evolving landscapes remains constrained by three key factors: tolerance thresholds of semantic feature matching under sudden terrain changes, storage-intensive processing requirements for onboard satellite deployment, and error propagation from public basemap quality discrepancies (cloud occlusion, resolution inconsistencies). Future research will prioritize three optimization pathways: 1) integrating real-time deformation monitoring data for dynamic landscape adaptation; 2) developing lightweight neural compression architectures to reduce storage overhead; and 3) implementing multisource geodata fusion with generative AI reconstruction and uncertainty-aware weighting mechanisms. These enhancements aim to extend the SEGPR method's environmental generalizability while maintaining refinement robustness through adaptive data synthesis.

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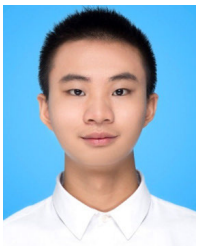
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